Argentina Poverty Update 2003

Background Paper No. 1

ARGENTINA'S CRISIS AND ITS IMPACT ON HOUSEHOLD WELFARE

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I. INTRODUCTION[1]

Argentina is currently in the midst of a deep and unparalleled economic, social, and political crisis. After three years of continuing recession, the economic and financial crisis deepened during 2001. Various attempts were made throughout 2001 to spur growth as a prelude to improved public finances and the debt profile but to no avail. Market perceptions of risk of default increased, especially in mid-2001. Efforts to protect the banking system from wide-spread withdrawal of deposits (the *corralito*) resulted in strangling of liquidity and economic activity and generated popular discontent. This finally culminated in the resignation of then President de la Rua in December 2001, a quick succession of appointed Presidents, a formal announcement of default on public sector borrowings, and in early 2002, the abandonment of the Convertibility Plan.

Now in its fourth year of recession, the economy this year is projected to decline by about 16 percent, bringing the decline in GDP to over 25 percent since its 1998 peak. With the peso depreciating sharply since its float and undefined monetary policy, Argentina is experiencing significant inflation for the first time since 1991. Between the decline in economic activity and the currency's depreciation, per capita GDP has declined sharply, to a projected US$2,850 in 2002 (down from US$8,210 at its peak in 1998).

The downturn of the past six months has sharply aggravated the already difficult social conditions in the country. Already high levels of unemployment (18.3% in October 2001) have jumped to 21.5 percent as of May 2002 according to official indicators.[2] Moreover, the destruction of jobs (particularly unskilled jobs) has been sharp. During the first quarter of 2002 the construction industry (key for unskilled labor) fell by 25 percent with respect to the last quarter of 2001 (42% with respect to the first quarter of 2001).[3] Poverty is also on the rise, confirming the wellknown links between economic crises and sharp increases in poverty (Lustig 2000). Official estimates for 28 urban centers indicate an increase from 38.3 % in October 2001 to 53.0 % in May 2002. This extremely negative context has also had a strong impact on the health and education sectors where there is growing evidence of deterioration in service delivery. The combined effect of all these factors has resulted in an increasingly conflictive social situation with high levels of crime, violence and diverse forms protests.

This paper summarizes the initial results of a household survey specially conducted to assess the effects the economic crisis has had on household welfare and identify the coping mechanisms and strategies adopted by Argentine families during these hard times. Specifically, the survey was designed to identify changes in income, consumption patterns, access to various social and other services, savings and payment patterns, labor market participation and reliance on formal and

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[2] INDEC, Encuesta Permanente de Hogares, total urban areas.

[3] Encuesta de Indicadores Laborales (EIL), Ministry of Labor.

informal safety net mechanisms. Additionally, the survey sought to assess changes in mental and emotional status of individuals.

The survey is nationally representative and covered 2,800 households in different regions.[4] Unlike most other surveys in Argentina (including INDEC's *Encuesta Permanente de Hogares*), this one included small localities (less than 2,000 inhabitants) and thus provides a unique perspective on rural areas. The survey, however, did not include dispersed areas and thus does not present a full picture of Argentina's heterogeneous rural sector. For urban areas, the sample design was based on a stratification by city size by region. For the rural areas, localities were selected randomly but, even though an effort was made to include localities from various regions, the sample is not large enough to be representative at the regional level. Overall, the survey design allows comparisons with the EPH for the urban areas.

The first part of the questionnaire asked for information for all household members and includes questions on demographic characteristics, employment, income, migration status, education level, health coverage and contributions to social security. For several of these questions, retrospective questions were asked using October 2001 as the reference point. Individuals were asked about their individual labor and non-labor incomes (cash and in-kind) as well as official and non-official income transfers.

The second part of the questionnaire focused on the household as a unit inquiring the use of savings, changes in consumption patterns, and participation in social programs and community activities. The questions were answered by the head of household (or the best informed individual in the household).[5] The questions on mental and emotional status and expectations for the future were asked only to this household member. The questionnaires for urban and rural areas differed slightly to capture specificities related to those areas. The field work was conducted during the months of June and July of 2002.

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[4] The survey was conducted by IBOPE.

[5] If the head of household did not know the answer to a particular question (e.g. changes in consumption patterns) the household was visited up to three times to find the member that could respond the specific question.

II. KEY FINDINGS

Household incomes. Table 1 presents the basic information on incomes derived from the survey. Monthly reported income per-capita from all sources amounted to $214.[6] This compares with an average income of $ 233 estimated by INDEC for urban areas for May 2002. Following standard procedures, per-capita incomes were adjusted using an adult-equivalency scale (to reflect differences in consumption needs) yielding an average monthly income of $264. We also present some standard inequality measures that confirm the relatively high levels of income inequality observed in the Argentine household surveys.

Table 1:
Mean income by quintiles- June 2002 - Total Country

Quintiles	*Adult-equivalent household income ($)*		*Per capita household income*	
	Mean ($)	Shares	Mean ($)	Shares
I	40.5	3.1	32.0	3.0
II	107.4	8.1	85.9	8.0
III	182.7	13.8	146.3	13.7
IV	300.1	22.7	240.9	22.5
V	690.8	52.3	564.7	52.8
Mean	264.3	100	214	100.0
Y10/Y1*	28.9		28.4	
Gini	0.473		0.478	

First quintile includes households reporting no income

* Y10/Y1 is the ratio of mean income corresponding to the highest and lowest deciles in the distribution. Y1 does not include those reporting zero income.

As mentioned before, a unique feature of this survey is the inclusion of rural areas that are not covered in the EPH. Table 2 confirms the well-known (but not necessarily well established) fact of higher urban incomes indicating a gap in the order of 60%.

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[6] Due to the known difficulties in obtaining truthful data on income levels, those persons that preferred not to report the exact amount of their earnings were requested to select one option between a menu of different ranges of income levels (and imputing the mid-point values). This allows us to reduce the number households with incomplete information on income from 17.2% to 7.6%. For those individuals reporting earnings who did not report specific amounts, we estimated earnings using Mincer equations whereby the logarithm of labor income per hour is defined as a function of the observable individual attributes such as sex, education, age and experience. To correct for selection bias we used the Heckman method. The absolute and relative effects on incomes of these corrections are insignificant (per capita income increases from $212 to $214).

Table 2:
Mean income by areas

Areas	*Adult-equivalent household income ($)*	*Per-capita household income ($)*
Urban	268.3	223.27
Rural	168.9	142.13

In order to identify the effect of the crisis on family incomes, the survey asked each household to indicate if its current income had changed –in nominal terms—relative to that corresponding to October 2001. Table 3 shows that over 40% of households reported no changes in income while almost 50% of the households reported a reduction in nominal incomes. Notably, about 8% of the households reported increases in incomes.

Table 3
Changes in incomes

With respect to October 2001	%
Increased	8.0
No change	43.4
Reduced	48.6

The information collected does not allow us to determine the extent to which these changes influenced the distribution of incomes given that the retrospective questions did not ask for income levels. Table 4, however, shows the incidence of changes according to the education level of the head of household –a proxy for household income. It is interesting to note that the highest incidence of reductions in income levels corresponds to households headed by individuals with secondary education -- a group that tends to be located in the middle of the income distribution ('the middle class') -- and the lowest incidence of reductions in income levels corresponds to households headed by individuals with incomplete primary education --a group that tends to be located at the lowest end of the income distribution ('the poorest'). This information suggests middle classes might have been the most seriously affected by the economic crisis[7].

[7] On the contrary, during the 1990's, economic shocks had a deeper effect on the poor and those with the lowest education levels. SeeWorld Bank, 2001.

Table 4:
Changes in income by education level of the head of household

	Increased	No change	Reduced	Total
Incomplete primary education	4.25	50.14	45.61	100
Primary education	7.81	44.85	47.34	100
Incomplete secondary education	15.44	34.52	50.04	100
Secondary education	6.72	39.51	53.77	100
Incomplete high education	6.83	45.94	47.23	100
High education	8.91	41.24	49.85	100
Total	8.07	43.22	48.71	100

Of particular interest is the case of the approximately 8% of the households reporting an increase in incomes with respect to October 2001 given the extreme negative economic performance during this period. Table 5 provides us with some clues as to the main reasons behind this increase in nominal incomes.

Table 5:
Households reporting increases in incomes

Reasons	%
Added workers	15.0
Changed jobs	14.1
Worked more hours	14.4
Same job-same hours	31.7
Higher non-labor incomes	22.3
Other	2.5

In almost half the cases, the reported increase in income is associated with a change in employment characteristics: a new worker in the family (a majority in temporary, informal jobs), a change to a better-paying job, or more hours worked. The other half of the cases do not show any noticeable change in employment characteristics for any of the household members. In a little over 20% of the cases it is possible to identify an increase in non-labor incomes as the principal source of change.

Employment. The survey also inquired about the labor market status of individuals as well as changes with respect to their employment status in October of 2001. The results indicate a stable rate of activity (i.e. no net change in labor force participation) in the neighborhood of 40%, with an increase in the unemployment rate in the order of 3% combined with a reduction in the employment rate (as a proportion of the population) of slightly more than 1%. A similar increase in the unemployment rate has been registered by the EPH survey (from 18.3% to 21.5%) for 28 urban centers even though with slightly different levels.

Table 6:
Employment (%)
May / June 2002

	May/June 2002	October 2001
Employment rate	29.9	31.1*
Activity rate	39.8	39.8*
Unemployment rate	24.9	22.0*
Unemployment rate (EPH)	21.5	18.3

(*) Based on retrospective information provided by the survey respondents

Of course, these aggregate numbers hide significant changes in labor market status for specific households. Table 7 provides a summary view of those changes by focusing on the change in employment status between October 2001 and the time of the survey based on individual recollections.

Table 7:
Changes in labor market status

			Previous status				
			Unemployed	Inactive	Permanent Job	Temporary Job	Total
Present status	Unemployed	*1000s*	1,435	1,228	15	555	3,233
		%	50.6	6.2	0.2	17.1	9.9
	Inactive	*1000s*	1,039	18,252	20	386	19,696
		%	36.7	92.5	0.3	11.9	60.0
	Permanent Job	*1000s*	113	91	6,498	331	7,034
		%	4.0	0.5	92.6	10.2	21.4
	Temporary Job	*1000s*	247	153	485	1,969	2,854
		%	8.7	0.8	6.9	60.8	8.7
	Total	*1000s*	2,834	19,725	7,018	3,241	32,817
		%	100	100	100	100	100

The stability in labor force participation is confirmed by comparing the numbers in the "inactive" category on both sides of the matrix. The number of individuals moving out of the labor force (mostly previously unemployed) is similar to that of individuals entering the labor force (and mostly remaining unemployed). A more detailed analysis shows these are mostly 'secondary workers' in households in which there is another member employed.

Not surprisingly, the number of individuals losing jobs exceeds that of individuals finding jobs. As Table 8 shows, the group of those moving out of unemployment tends to be composed by

secondary workers in a higher proportion than the group of those losing jobs -- indicating the presence of some changes in family roles to be further analyzed.

Table 8:
Groups that suffer changes classified by household member and gender

	Unemployed to employed (%)	*Employed to unemployed (%)*
Household member		
Head of household	35.9	57.0
Spouse	18.8	12.6
Others	45.3	30.4
Total	100.0	100.0
Gender		
Male	58.4	62.0
Female	41.6	38.0
Total	100.0	100.0

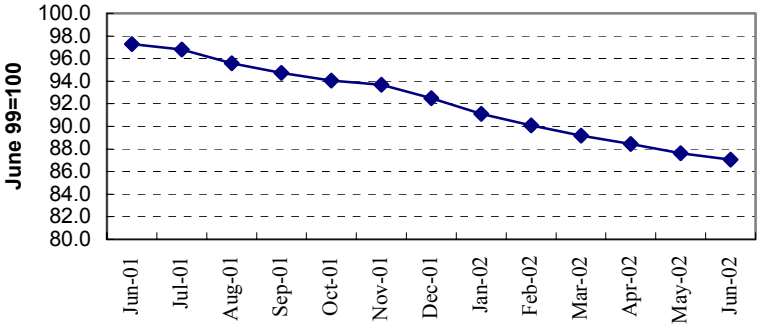
The third fact worth mentioning is an apparent deterioration in the 'quality' of jobs. This is noticeable in terms of the type of employment reported both by those that obtained new jobs (temporary more than permanent, and within those with permanent jobs more than half do not report any type of the standard benefits associated with formal sector jobs), and those that changed jobs (indicating a net increase in the proportion of temporary jobs and a large presence of 'jobs without benefits' among those that moved to permanent positions). Approximately 6% of those employed reported losing at least some type of benefit. This trend is consistent with the one observed through the EPH.[8]

The sharp deterioration in formal sector employment is confirmed independently by the Ministry of Labor's "Encuesta de Indicadores Laborales" (EIL) which covers more than 1,500 firms representing a universe of approximately 38,000 firms of various sizes and in different sectors employing approximately 1,9 million registered workers in the three largest urban centers (Greater Buenos Aires- Greater Rosario and Greater Cordoba). As shown in Graph 1, formal employment is estimated to have fallen by 7.4% since October 2001, and 10.5 % since June 2001[9].

[8] For example, the proportion of salaried workers without benefits increase from 33 to 35% between the October 2001 and May 2002 EPH.

[9] Additional data supplied by AFIP (Administración Federal de Ingresos Publicos) reveals similar results: the monthly average of jobs in the first quarter of 2002 shows a decrease of 10,1% compared to the same period of the preceding year.

Graph 1: Evolution of formal employment
(Greater Buenos Aires)



Source: EIL

Poverty. Based on the information on incomes corresponding to the month of June, we have estimated the incidence of poverty using four different lines. First, using the official 'poverty' and 'indigence' lines estimated by INDEC for the month of June 2002.[10] . For rural areas, we used the poverty line corresponding to the urban center in the same region. This, of course, does not recognize the potential differences in urban/rural prices but at least controls for regional differences (See World Bank 2000)[11] In addition we estimated a 1-dollar and 2-dollar-a-day poverty line using the standard World Bank methodology for the entire country.[12] The results for the country are shown in Table 9 while Table 10 presents results for urban and rural areas.

Table 9:
Poverty Rates June 2002-Total Country
(% of individuals)

	WB-Survey	EPH	
	June 02	May 02	Oct 01
Poverty	53.7	53.0	38.3
Indigence	23.8	24.8	13.6
PPP (U$1)	9.9	9.5	7.5
PPP (U$2)	24.6	20.8	17.1

[10] The basic food baskets were originally estimated for each urban center based on consumption patterns for 1997 and were adjusted to reflect prices prevailing in April 2001 in each center using INDEC's "Paridad de Precios de Compra del Consumidor (PPCC)" index for that month. The lines were further updated to the month of June using the variation in prices in Great Buenos Aires. The absence of systematic price information for all urban centers is a serious limitation in that sense. We compared changes in prices of food-stuff in Buenos Aires, Córdoba, Mendoza and Tucumán during the first six months of 2002 which indicate a slighter higher rate of increase in the interior than in GBA.

[11] A more in-depth analysis of poverty in rural areas will be conducted separately.

[12] We constructed the poverty lines using the PPP for Argentina in 1993 (PPP= 0.7779) and converting to local prices using the CPI.

Table 10:
Poverty Rates June 2002-by areas

	Urban (% of individuals)	*Rural (% of individuals)*
Poverty	52.9	72.6
Indigence	23.2	40.1

A few points are worth mentioning. First, the incidence of poverty/indigence (using official lines) confirms the predictions made at the early stages of the crisis that suggested poverty rates in the order of 50% and indigence rates in the order of 25%. Official estimates from EPH reveal similar poverty and indigence rates (53% and 24.8% respectively), although with a more reduced coverage given its exclusion of rural areas. Second, with all the necessary caveats in mind, it is important to note that the 2-dollar-a-day poverty line is very close to the official indigence line which, in principle, provides a better sense of how poverty in the 'post-default' Argentina compares in international terms. Third, and once again with all the necessary methodological caveats, even though the incidence of poverty/indigence is, as expected, much higher in rural areas, the addition of rural areas in the estimation does not have a significant effect on national rates given the relative size of the two groups.

Subjective welfare. With the exploratory purpose of identifying changes in the mental and emotional state of individuals ('how people feel'), the survey asked the key respondent in each household whether they 'felt discouraged' (*se siente desanimado?*) and/or 'hopeless about the future' (*desesperanzado con respecto al futuro*). Individuals were asked about the main reason for those feelings and how they felt back in October 2001.

Table 11:
Changes in emotional status

	Feesl Discouraged		*Feels hopeless about the future*	
	Now (%)	Before October 2001 (%)	Now (%)	Before October 2001 (%)
Never	26.8	47.8	21.1	45.5
Sometimes	34.2	37.3	33.2	36.2
Frequently	24.8	9.6	27.5	11.3
Always	13.5	4.0	16.7	4.8
No response	0.6	1.3	1.5	2.2
Total	100	100	100	100

As shown in Table 11, on both dimensions the survey indicates a high and increasing incidence of discouragement and pessimism among respondents. Table 12 provides further evidence on the source of this discouragement/pessimism among those reporting such feelings. The preeminence of 'economic reasons' is simply overwhelming for both dimensions.

Table 12:
Reasons reported for emotional status

	Feels Discouraged (%)	*Feels hopeless about the future (%)*
Does not find a job	11.3	12.7
Difficulties in the job	1.6	1.6
Economic reasons	53.4	54.0
Health reasons	4.8	3.0
Something difficult to explain	16.1	15.4
Other reasons	12.8	13.4
Total	100.0	100

Among those reporting feeling discouraged now but not before October 2001, we sought to identify whether this change was associated to the employment status of the respondent. As shown in Table 13, this does not appear to be the case. Similarly, we didn't find any pattern across the income distribution. These findings suggests a generalized impact of the economic crisis on the emotional status of the population.

Table 13:
Employment status of those reporting increasing discouragement

Present status	*%*
Employed	46.4
Unemployed	14.5
Retired	15.1
Student	3.4
Housewife	10.4
Others inactive	10.1
Total	100.0

Social unrest. The survey provides confirmation of a sizeable increase in the participation of individuals in various forms of social protest –up from 7.6% before October 2001 to 16.2%. Interestingly, participation increases as one moves along the income distribution. This is consistent with another result of the survey: the most popular form of protest was the *cacerolazo* or pot-banging (9.2% participated) – a form of protest that became extremely popular in the major urban centers during the first quarter of 2002.[13]

[13] Lower income groups participated mostly in *piquetes* or road closings.

Table 14: Participation in social protests
(% of households participating)

Quintiles	I	II	III	IV	V	Total	Total Before Oct. 2001
%	11.4	12.6	17.5	17.5	22.0	16.2	7.6

Violence appears to be another important factor in the post-crisis period. As shown in Table 15, 20% of the households reported being a victim of crime or violence in the previous six months. It is interesting to notice that while it is true that the highest income group is the one reporting the highest incidence of crime of violence, there isn't a simple correlation between victimization and level of income. The data is consistent with the information collected by the Direccion Nacional de Politica Criminal for Gran Buenos Aires, Rosario and Cordoba which indicates that during 2000 13% of those interviewed had been victim of some type of crime. The information does not find significant differences incidence among socio-economic groups but does find that higher income groups tend to report more the occurrence of such events (particularly due to the robbery of cars).

Table 15: Exposure to crime and violence
(% of households)

Quintiles	I	II	III	IV	V	Total
%	19.7	22.9	14.5	19.9	25.3	20.4

III. COPING STRATEGIES

In this third section of the paper, we focus on strategies adopted by households to cope with the crisis. We first provide a general description of the key types of strategies followed by households. We have grouped them in three categories: (i) *adaptative household strategies*; (ii) *active household strategies* and (iii) *social network strategies.* We then analyze the components of each category and identify patterns whereby some strategies tend to be followed by different types of households (e.g. by income group). This is as a first, exploratory, step in our ongoing analysis of coping strategies.

A general view. As indicated before, the survey inquired about the most frequently strategies used by households in reaction to the worsening economic conditions experienced over the previous eight months.

The first set of strategies, which we grouped under the title of *adaptative household strategies*, refers to those households that responded to the crisis by changing consumption patterns including the use of various types of services. The second set of strategies, which we grouped under the title of *active household strategies*, involve the incremental use of physical, financial and human assets available to the household. These strategies include: adding new workers, working more hours, selling assets, using savings, borrowing, migration. The last group of strategies, s*ocial network strategies,* includes strategies that rely on assistance from friends, family, NGOs or Government.

Table 16 summarizes the key information regarding the use of these different strategies for all families and specifically for the group of families who experienced a reduction in household income.

A first aspect to notice is that almost all households, regardless of whether they experienced changes incomes, reported having changed their consumption patterns in response to economic crisis. This is true across the income distribution (see Table 17). Argentine families have, indeed, adapted to the economic crisis by changing their consumption patterns.

Table 16 :
Use of strategies by households

	% of household using:	% of households reporting a reduction in income using:
(i) Adaptative household strategies	98.2	99.1
(ii) Active household strategies	37.3	45.0
(iii) Social Network Strategies	32.7	36.4
Do not use any of these strategies	1.5	0.8
At least one of them	98.5	99.2
At least two of them	54.7	61.6
All of them	13.6	17.8

Furthermore, with the exception of active strategies that tended to be used more intensively by household suffering reductions in income, no noticeable pattern is observed between these two groups when considering the use of those broad set of coping strategies. However, when attention is focused on income groups (Table 17), regardless of whether or not they experienced reductions in income, it becomes apparent that the use of social network strategies is strongly correlated with income levels.

Table 17:
Use of strategies by income quintiles[14]
(% of households in the income groups using:)

	I	II	III	IV	V	Total
(i) Adaptative household strategies	99.8	98.1	98.9	97.3	96.6	98.2
(ii) Active household strategies	47.0	37.7	36.1	30.5	30.7	36.4
(iii) Social Network Strategies	55.5	39.1	38.4	24.2	12.0	33.8

As we will show next, beyond these general patterns, significant differences between various groups are found once the broad categories are further desegregated.

Adaptative household strategies. Table 18 summarizes the type of changes adopted by families in their expenditure patterns. In particular, we focused on both reductions in consumption levels and substitution for less expensive items and distinguished between essential (food) and non-essential (other) goods.

Overall we see a picture whereby all families have reacted to the economic crisis through a combination of consuming less and substituting for less expensive goods both for essential and non-essential items.[15] Having said this, two patterns are observable. First, higher income groups appear, not surprisingly, to have avoided to a greater extent such changes. This is particularly true in the case of food where the difference between income quintiles is striking. Second, the substitution for cheaper products (or second-hand products when available) appears to be the more pervasive than reductions in consumption levels. In other words, the information suggests that families have tried to shield the consumption of food as much as possible reducing the consumption of other goods and switching to less expensive products. The lower income groups, not surprisingly, appear to have been less successful in doing so.

[14] Quintiles in this section are constructed with same quantity of households in each quintile, sorted by familiar income adjusted by household size (using adult equivalent), excluding households with incomplete reported income.

[15] The CPI increased by 30% during the first six months of the current year. The increase in the prices of food items was higher (39%) than average. Using the expenditure shares for the lowest quintile derived from the 1997 consumption survey we estimate the cost of living for the poorest groups in society increased 45% during this period.

Table 18:
Use of adaptative strategies by income quintiles

Changes in consumption patterns	I	II	III	IV	V	Total
Food						
Reduced consumption	90.4	83.1	73.2	69.0	59.1	74.9
Substituted for cheaper products	97.6	95.4	92.5	91.5	84.8	92.3
Other goods						
Reduced consumption	90.5	87.7	81.5	76.8	68.3	81.0
Substituted for cheaper products	89.5	89.3	80.4	80.2	76.6	83.2
Purchased second hand products	51.7	40.2	34.8	33.0	24.3	36.7
Dismissed/reduced the use domestic services*	.	.	43.0	27.8	40.5	37.9

*Estimated only for those households for which this was an option and only when a sufficient number of cases were observed.

The survey also allows us to observe changes in the use/access to various types of services. Here we are focusing on education, health and public services.

A question that deserves particular attention –and for which we are conducting a separate analysis using various sources of information—is whether there is any evidence of increased school drop-out as a result of the economic crisis. The survey provides some initial hints indicating limited evidence of such an effect. Table 19 looks at three age groups: 6-12, 13-15 and 16-18. The second column provides rough estimates of coverage rates for each group –without any reference to the level of schooling. With all its limitations, this data confirms the rates traditionally observed through household surveys and does not show signs of reductions in coverage rates.

We have further analyzed whether there are any indications of drop-outs during the first months of the current school year among those children reported not to be at school (that is, those not included in column two of the table). Household heads were asked whether someone in the household (without identifying who it was[16]) had either dropped out or postponed starting school. The third column reports the percentage of children in the age group that are not in school and belong to a family in which there is at least one member that dropped out or postponed starting school. Of course, this might be over-estimating the presence of drop outs to the extent that there is more than one household member that had dropped out or postponed starting school (including higher education) and in that sense it is an estimate of the maximum potential size of the effect. The first thing to notice is that, given the very small number of cases involved, the estimates are not sufficiently precise (i.e. have a high coefficient of variation).

[16] This avoided a sort of 'finger-pointing' effect that might have led to under-reporting of cases of drop outs at the expense of losing information on exactly which member of the household was involved.

Table 19:
Have families taken children out of school?

	Goes to school (% of all in group)			Drop-out estimate
Age group	Total areas	Urban area	Rural area	Total areas
6 to 12 years	97.0	97.3	92.1	0.2
13 to 15 years	92.5	92.5	90.9	0.6
16 to 18 years	80.2	81.1	60.1	2.3

Taking this into account, it is apparent that there is no evidence of drop-out for children in the 6-12 or 13-15 age bracket and the potential for a small (but statistically non-significant) presence of drop-outs in the 16-18 group.

As a consistency test we have checked, for the latter group, their location within the distribution of household incomes. As shown in Table 20 the few cases of apparent drop outs belong mostly to the lowest deciles of the distribution offering some additional level of comfort regarding the evidence.

Table 20: Potential school drop-outs

Quintiles	"Drop-outs in the 16-18 group (%)
I	53.9
II	23.3
III	6.3
IV	16.6
V	0

The limited evidence on children dropping out of school as a result of the crisis could indicate that families try very hard to protect what is perhaps the most important type of investment they can make on their children even under difficult circumstances (De Ferranti et al., 2000). However, as evidence from other crises suggests, the negative effects might be felt not immediately but several months after the beginning of the crisis (e.g. might be felt in enrollment rates in 2003). Furthermore, the data from the survey does not allow us to identify the number of children not attending school on a regular basis, perhaps a more prevalent type of adjustment in many cases.

Table 21 provides some further evidence on whether families made adjustments that affected their children's education. The share of families that substituted schools (to less expensive private schools or from private to public schools) is very small, suggesting a resistance to make such changes even in the context of a sharp economic deterioration.

On the other hand, a very large proportion of the households (particularly in the lower income groups)[17] reported reducing the purchase of schools materials –with yet unclear effects on the

[17] Almost 90% of the households in the lower income quintile reported reductions in the purchase of school materials while only 43% of those in the highest income reported the same.

quality of schooling. The difficulties experienced by several provinces in keeping up with spending levels suggest the presence of restrictions on the supply side that may be more serious than those on the demand side, at least at this early stage of the process.

Table 21:
Adjustments in education expenditures
(% of households that)

Substituted private school for public one	2.0
Turned to cheaper private schools	3.1
Reduced purchase of school materials	71.9

In the case of health services, there is evidence of severe negative impacts since early on in the year both in terms of difficulties within the social health insurance and public health sub-systems. The information collected through the survey confirms the trends identified previously.

Approximately 12% of the individuals experienced some change in health insurance coverage (see Table 22). More than 60% of those lost coverage altogether –mostly concentrated in the lowest income groups. For the approximately 40% of the individuals experiencing a change in health insurance coverage a movement toward lesser coverage is observed. Almost half those that previously had double insurance coverage (*obra social* and *pre-paga*) now only maintain their primary (*obra*) health insurance. A shift from full private insurance to just emergency insurance, is also observed.

Table 22: Changes in health insurance coverage
(% of those individuals experiencing a change in coverage)

Quintiles	I	II	III	IV	V	Total
Lost all form of coverage	76.0	61.1	78.6	52.6	33.6	61.4
Changed coverage	24.0	38.9	21.4	47.4	66.4	38.7

The loss of insurance coverage is leading an increasing number of people to rely on the use public health facilities. Of those households reporting to have substituted private for public health facilities as a reaction to the economic crisis, 60% have experienced some form of loss or limitation in the coverage offered by their health insurers. More specifically, 16% lost their insurance altogether, 5% have outstanding bills that do not allow them to use their insurance coverage, 13% cannot afford paying the copayments, 8% experienced a reduction in the services covered by the insurers and 13% are retirees experiencing difficulties with the services offered by PAMI.[18]

Overall, the survey found that families have, in one way or another, cut down on health services as a result of the economic crisis. A few results are worth mentioning. Almost 23% of the households report that at least one member experienced lack of access to health services. Three fourths of

[18] PAMI is the national health insurer for retirees.

those report the reason was "lack of money" to pay for medicines (44%), transportation costs (26%) and fees (5%). More specifically, 37% of the households with children under the age of twelve reported having reduced the frequency with which they take their children for medical checkups. Almost 45% of those belong to the lowest quintile of the income distribution. But this effect was also felt in the second (29%) and third (18%) deciles.

Cutting down on the use of utility services appears as another strategy followed by households. Typically, this implies either stopping/delaying payments (with the risk of being cut-off) or directly asking for the interruption of the service. Between 30 and 50% of the households report delays in payments. The percentage of households reporting being cut-off from various services (electricity, gas, water, telephone, cable, internet) range from 2% (in the case of water supply) to 14% for cable. In most cases, the key reason has been lack of payment.

Similarly, households have tended to substitute more for less expensive means of transportation, as shown in Table 23. More than half the households previously using only public transportation, now report having increased the use of a bicycle or simply walking. A similar switch from cars and taxis to public transportation is observed as well.

Table 23: Changes in the use of transport services

	I	II	III	IV	V	Total
Car/taxis by public transportation	57.4	56.7	47.8	45.5	32.4	46.1
Car/taxi/ public transportation by bicycle/walking	73.9	62.5	61.3	49.0	33.5	55.4

These figures are consistent with those emerging from the official data on the use of public services[19] that indicate, for the first five months of the year, a 22% reduction in the number of passengers in urban railways, a 9% in subways and 14% metropolitan bus services.

Active household strategies. Table 24 summarizes some of the active measures households have adopted in response to the economic crisis. In approximately 13% of the households we observed the entry into the labor force of a new member. This is particularly strong among the lowest income group (Table 25). It is typically a secondary worker: in 50% of the cases it is the son or daughter and in 25% of the cases the spouse of the person reported to be the head of the household. The effectiveness of this coping strategy, however, appears to be rather limited. A little over 13% of those new entrants succeed in finding a new job. Those succeeding tend not to be the poorest among the entrants.

[19] These statistics are collected monthly by INDEC and constitute the basis for its Indicador Sintetico de Servicios Publicos (ISSP).

Table 24 : Active household strategies

	% of household using	% of households reporting a reduction in income using:
Labor market strategies		
Adding new workers to labor market	13.4	16.1
Working more hours	14.8	19.2
Time		
Increased home manufacture	59.9	62.6
Dismiss/reduce domestic services*	35.3	40.4
Migration	4.1	3.9

*Percentage for this item refers to he households for which the category is relevant

Working more hours appears as another strategy followed by all income groups. However, it tends to be mostly salaried workers in permanent jobs with a particular concentration in commerce that appear to succeed in this effort to increase the hours of work.

In parallel to these changes in employment strategies, it is possible to observe a tendency to substitute leisure for in-home work as a way of coping with the crisis. Two examples are worth mentioning. First, approximately 60% of the households report they have increased the time allocated to the preparation of 'home-produced-goods' (e.g. food) to substitute for purchased goods. As shown in Table 25, this strategy is heavily used by the lowest income groups. Secondly, among those household employing domestic service, there is a noticeable reduction in its use.

Table 25 : Active household strategies by income level

Quintiles	I	II	III	IV	V	Total
Labor market strategies						
Adding new workers to labor market	28.0	16.8	12.2	6.2	1.4	12.9
Working more hours	11.4	15.6	16.3	11.5	13.4	13.7
Time						
Increased home manufacture	74.4	73.0	62.6	52.5	43.2	61.1
Dismiss/reduce domestic services	.	.	43.0	27.8	40.5	37.9
Migration	4.3	2.3	6.0	5.1	4.6	4.5

The survey also inquired as to whether a household member had migrated since October 2001 or if someone in the household is considering that option. On average a little over 4% of the households reported having at least one member that migrated, a percentage distributed more or less evenly across the income distribution.[20] The main reasons for migration included 'lack of work' (58%) and seeking better quality of life (15%). On the other hand, more than 20% of the

[20] This excludes those migrating as a result of marriage, sickness or other family reasons.

households report that at least one family member is considering migrating. The key option being considered (80% of the households) is migrating to another country –particularly within the highest income groups in the City of Buenos Aires.

Since the beginning of the crisis a larger proportion of the households are using their savings, selling assets or borrowing as a means of smoothing consumption (Table 26).

Table 26: Use of financial strategies

	% of household using		% of households reporting a reduction in income using:	
	Now	Before October 2001	Now	Before October 2001
Selling assets	3.7	1.1	5.5	1.0
Using savings	5.1	3.0	7.1	4.9
Borrowing from banks	1.7	2.4	1.3	3.1
Purchase with delayed payment	7.3	5.1	9.7	6.6

Perhaps the most interesting fact to notice is how low income households rely as a coping strategy on the use of informal credit by neighborhood stores (*compra al fiado*) whereby payment is delayed (Table 27).

Table 27: Use of financial strategies by income level

Quintiles	I	II	III	IV	V	Total
Selling assets	5.9	3.7	3.3	2.7	1.1	3.3
Using savings	2.8	3.5	4.0	8.0	5.6	4.8
Borrowing from banks	0.9	3.6	1.8	0.6	2.0	1.8
Purchase with delayed payment	14.6	13.1	9.5	2.3	0.7	8.0

Social network strategies. Table 28 summarizes some of the key indicators of use of social networks by households. Table 29 provides further information by disaggregating by income groups. Overall, we observe that more than one third of the households report benefiting at least in one aspect from access to a network of social support. The use of such strategies is somewhat higher for households that experienced negative nominal shocks in their incomes (Table 28) and significantly higher for households in the lower end of the income distribution (Table 29).

Table 28: Use of Social Network Strategies

	% of household using	% of households reporting a reduction in income using:
Assistance from other persons who don't live in the house[21]	16.3	17.0
Borrowing from friends or family	10.7	13.7
Social Plans	6.9	7.5
Participation in community activities[22]	20.9	26.4
Barter	11.1	15.2
Using at least one	37.0	42.5
Using more than one	14.6	19.9

The survey provides some evidence that social networks might be cutting across the income distribution. Indeed, by comparing the distribution of those households reporting having received or given assistance to persons not living in their own household, we notice how households in the lowest quintile tend to be net recipients and those in the highest quintile net givers.

Table 29: Use of Social Network Strategies by Income Level

	I	II	III	IV	V	Total
Assistance from other persons who don't live in the house	26.4	15.5	20.9	15.8	7.3	17.1
Assistance to other persons who don't live in the house	10.0	19.7	17.1	21.2	37.9	21.2
Borrowing from friends or family	21.2	15.7	10.6	5.8	3.0	11.3
Social Plans	18.5	6.1	7.5	1.3	0.1	6.7
Participation in community activities	29.4	23.6	22.8	15.7	15.7	21.4
Barter	20.2	15.4	11.7	5.5	3.9	11.3

Through retrospective questions, we were able to identify changes in the access and/or use of social networks since October 2001. While the survey registers an increase in the use of various forms of social network use, the largest increases are in the use of barter and participation in community activities.

[21] A higher percentage of households (22%) report providing assistance to persons not living in the same households.

[22] Community activities include the following categories: school canteens, soup-kitchens, communal purchases of goods, community workshops, fund-raising, community child-care, among various others.

Table 30: Changes in the use of social networks

	% of household using	(Before Oct 2001) % of household using
Assistance from other persons who don't live in the house	16.3	13.2
Borrowing from friends or family	10.7	7.0
Assistance from Govt or NGOs	6.9	2.7
Participation in community activities	20.9	11.1
Barter	11.0	3.2

The intensity of use of barter among low income groups is documented by the fact that the per-capita income of households using barter is less than entire population ($143 vs. $214, and $183 vs. $264 with and without adjustment for adult equivalency respectively). Households using barter tend to have more unemployed members (43.3% vs. 22.4%) and more temporary workers (42% vs. 23%) –further proof of the higher vulnerability of those households.

A similar pattern is observed for those households that report receiving some type of official help. Their per-capita income is below average ($122.7 and $97.62 with and without adjustment for adult equivalent), have a higher incidence of unemployment (65% have an unemployed member) and, for those that are employed, they tend to have mostly temporary jobs (80%).

IV. BY WAY OF CONCLUSION

The paper presents compelling evidence of the magnitude of the social costs of the Argentine economic crisis with strong negative welfare effects throughout the income distribution. Household level data confirms the negative picture obtained from the analysis of macro level indicators available since early on in the year.

Households are using a variety of coping strategies to respond to a crisis of such a large intensity. The limited data collected on consumption does not allow us to determine to what extent the reductions in income levels are resulting in extreme forms of deprivation (i.e. to what extent the coping strategies are allowing families to avoid deep deprivation). However, the magnitude of the shocks and the limitations in the effectiveness of various coping mechanisms we have already been able to identify, suggest the effects on welfare are indeed very serious. The evidence suggests stronger effects on the use of health than education services –a trend we are analyzing in more detail using other sources of information.

A quick comparison with evidence from other countries experiencing similar large economic shocks, shows some very interesting similarities as well as some specific differences. The analysis by Fallon and Lucas (2002) on the effects on labor markets[23] indicates a pattern of rising open unemployment rates and important changes in the sectoral composition of employment (sharp reductions in construction and manufacturing --the latter as the corporate sector is hit by the higher costs of imported materials, difficulties obtaining credit and the rising burden of debt in foreign currency) similar to that observed in the case of the post-convertibility Argentina.

The evidence on the distributional impact of crises shows some differences between countries. Thomas et al. (1999) estimated that in the case of Indonesia there was a larger decline in per capita expenditures both the top and bottom quartiles of the distribution that for households in the middle of the distribution. In contrast, there is evidence of at least a weak redistribution of income from the middle classes to the rich in Thailand. Our estimates suggests the reductions in family income in Argentina affected the most the middle class.

Regarding coping strategies, as in the case of Argentina, Frankengerg et al. (1999) note that informal assistance from friends and family members were particularly important in Indonesia during the crisis, with about a quarter of the households receiving such assistance. Its median value was considerably higher than that of assistance from formal services. Sudarno, Wetterberg and Pritchett (1999), also found for the Indonesian case differences in coping strategies along the income distribution. For instance, middle-class families responding by working more hours, reducing consumption, drawing down savings and selling assets and lower income households resorting to more drastic measures, such as taking children out of school.

Regarding this last point, the evidence on the matter is mixed. For example, in Mexico during the 1982 crisis, high school dropout rates increased slightly while primary school dropout rates fell. Both changes were part of long term trends throughout the 1980s (Lusting 1998). Similarly, Adam and Chamberlin (1999) did not find evidence of significant reductions in school participation in Thailand even though they also recognized the measurement difficulties and delayed responses we

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[23] Countries studied and crisis years: Indonesia (1998), Korea (1998), Malaysia (1998), Thailand (1997), Argentina (1995), Mexico (1995) and Turkey (1994).

mentioned above. The evidence for Thailand, though, did find a gap in dropout rates between poor and non-poor households, as we did for the case of Argentina.

Finally, even though the magnitude of the current crisis is much larger than those experienced during the 1990's, the general characteristics of the coping strategies we observe are very similar to those observed in previous studies for Argentina (World Bank 2001). In particular, the reliance on social networks and the changes in consumption patterns.

Further steps in our ongoing analysis include more detailed analyses of the impact of the crisis on health and education services, a more systematic analysis of the use of various coping strategies by type of households (using multivariate methods), a more specific consideration of the rural sector – making use of the unique coverage of this survey--, and a particular focus on the role of migration as a coping strategy.

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